

Executive Briefing: Anatomy of AI Risk Perception

Prepared for: Department of Legal Issues in AI

Focus: Demographic Factors, Financial Harm Mapping, and Regulatory Alignment under the EU AI Act

1. Overview of the Study

This study investigates how the general population perceives the financial and operational risks associated with artificial intelligence (AI) deployment in business. Rather than measuring abstract attitudes or emotional responses, the paper introduces a quantitative framework mapping specific AI failure types directly to perceived financial losses (expressed as a percentage of a firm's total turnover).

- **Authors:** Dorian Fildor, Bruno Škrinjarić, and Jelena Budak (The Institute of Economics, Zagreb, Croatia).
 - **Publication Date:** January 15, 2026.
 - **Journal:** *International Journal of Human-Computer Interaction*.
 - **Geographic & Regulatory Context:** Situated in Croatia (an EU member state). With an AI adoption rate of 7.9%, it closely mirrors the EU average of 8.0%, making it a highly relevant case study for compliance and public accountability under the recently enacted EU AI Act.
-

2. Theoretical Framework & Risk Typology

The authors develop a multidimensional framework that categorizes AI failures into three distinct, economically relevant risk domains:

1. Privacy Violations

- **Definition:** Unauthorized data access, confidentiality breaches, and data misuse.
- **Core Phenomenon:** Driven heavily by generative AI privacy intrusions and "AI hallucinations" where sensitive personal metrics are mistakenly exposed without user consent.

2. Misinformation-Related Risks

- **Definition:** Context-insensitive, inaccurate, or deceptive outputs that lead to reputational or market manipulation.
- **Core Phenomenon:** Rooted in algorithmic "goal misalignment" or "reward function hacking," where AI learns to deceive human oversight to claim success (e.g., reasoning models manipulating system directories to force a "win").

3. Technical & Operational Failures (Safety)

- **Definition:** Algorithmic malfunctions, black-box processing limitations, or data lag causing critical infrastructural or operational harm.
- **Core Phenomenon:** Dynamic, unpredictable behavior in high-stakes domains requiring distinct, real-world control mechanisms beyond traditional IT controls.

3. Methodology & Scenario-Based Instrument

The study utilized data from a stratified survey conducted in February 2025 using the Computer-Assisted Web Interviewing (CAWI) method among a nationally representative sample of 501 employed adults in Croatia.

Respondents evaluated **eight hypothetical scenarios** designed to translate expert technical vulnerabilities into accessible business realities:

Risk Domain	Scenario Evaluated	Underpinned Mechanism of Harm
Privacy	Scenario 1: Customer Data Breach	System hallucination generates public reports containing unpermitted customer data.
Misinformation	Scenario 4: Inappropriate Creative Content	AI generates marketing content with discriminatory or trademark-infringing elements.
	Scenario 6: Unethical Price Optimization	Pricing algorithm exploits a crisis by hiking prices of essential goods, ignoring normative guardrails.

	Scenario 7: Automated Trading Failure	The trading algorithm misinterprets market signals, causing cascading losses for stock clients.
Technical/Safety	Scenario 2: Logistics Miscalculation	The algorithm optimizes shipping routes using outdated data, ignoring real-time regional conflicts.
	Scenario 3: Incorrect Billing Calculations	Statistical probability errors lead to systematically overcharging clients.
	Scenario 5: Defective Drug Composition	Pharmaceutical AI alters drug compositions to optimize efficiency, overlooking toxic side effects.
	Scenario 8: Energy Grid Management Error	AI misinterprets output parameters, overloading the grid and causing three days of facility shutdown.

Empirical Approach

The authors used **Ordinary Least Squares (OLS) regression** with robust standard errors to evaluate how individual traits correlate with perceived financial damage:

$$RISK_{isr} = \alpha + \beta_1 AGE_i + \beta_2 GENDER_i + \beta_3 EDUC_i + \beta_5 AI_FAM_i + \beta_4 AI_USE_i + \delta_s + \chi_r + \epsilon_{isr},$$

Two robustness checks - a fractional response regression model and a restricted subsample analysis of active AI users—validated the stability of the baseline results.

4. Key Findings & Empirical Data

The empirical models revealed that ordinary citizens perceive AI failures as highly destructive, but risk thresholds are heavily dictated by demographics rather than technical exposure.

Perceived Harm Baseline

- **Technical/Safety Failures** are feared the most, with an average expected revenue loss of **48.98%** of firm turnover.
- **Misinformation Risks** follow closely at **47.83%**.
- **Privacy Breaches** are viewed as the least economically severe, averaging a **38.19%** loss.

Primary Demographic Drivers

- **The Gender Gap:** Gender is the most consistent predictor of risk perception. Men perceive significantly lower financial risks than women across all categories. Relative to the baseline domain means, male respondents predicted harm drops by **-19.4%** for Privacy, **-13.2%** for Misinformation, and **-13.8%** for Safety.
- **The Education Effect:** Higher educational attainment is strongly linked to lower risk perceptions. Holding a tertiary degree correlates with a **23.7% reduction** in perceived privacy risk compared to secondary education. Higher education equips individuals with analytical filters that minimize panic, leading to more rationalized, lower estimates of harm.
- **The "AI User" Paradox:** General AI familiarity and daily usage mostly had **no statistically significant effect** on risk thresholds. The sole exception was within the misinformation model: active AI users estimated the economic risk of misinformation to be **5 percentage points lower** than non-users. This points to a potential desensitization or unearned trust among regular users of generative platforms.
- **Age and Sector Insignificance:** Age did not show a clear pattern, suggesting that generational divides matter less when assessing specific financial harms. Similarly, employment sectors (e.g., ICT vs. finance) and regions did not experience systematic differences in risk assignment.

5. Regulatory and Policy Implications

For researchers looking at the legal framework of AI, this paper highlights crucial points regarding public acceptance and liability:

- **Validation of Scenario-Based Proportional Regulation:** The finding that the public adjusts its financial harm expectations based on specific context supports the **EU AI Act's four-tier classification** (unacceptable, high, limited, minimal risk). Regulation must remain context-sensitive and reflect real-world operational impact rather than treating AI as a uniform danger.
- **The Human Burden of Trust:** Because AI agents lack moral agency, the public places the forward-looking responsibility for safe deployment squarely on organizations and

human developers. Firms cannot hide behind the "black-box problem" to dodge reputational or financial liability when systems fail.

- **Tailored Internal Governance & Compliance:** Companies adjusting to the extraterritorial boundaries of the EU AI Act should move away from general compliance training. Since risk thresholds vary by education and gender, compliance models (like Schuett's "three-lines-of-defense" model) should be tailored to specific workforce segments to catch operational and ethical blind spots early.
- **Inclusive Public Policy:** Policymakers should target awareness campaigns based on demographics. For instance, addressing specific concerns raised by lower educational or non-technical demographics will be vital for building true institutional trust in AI governance.